

Lab # 13a

Shell Scripting Using Switch Case

I. The case Statement

The case statement checks the value of a variable against a large number of possibilities. Unlike, case does not check expressions it only matches strings.

```
case word in

    pattern1)

        Statement(s) to be executed if pattern1 matches

    ;;

    pattern2)

        Statement(s) to be executed if pattern2 matches

    ;;

    pattern3)

        Statement(s) to be executed if pattern3 matches

    ;;

esac
```

```
case value in pattern1)
    commands
;;
pattern2)
    commands
;;
.
.
esac
```

Some things to note about the above syntax

- The case statement begins with the word case followed by a value and then the word in. Value is typically a script argument or other shell variable.
- The next line contains a pattern followed by a closing parenthesis. There is no opening parenthesis and there should not be a space between the pattern and parenthesis. The pattern can be any of the following.
- The word esac (case spelled backwards) ends a case statement.

Example 1

```
#!/bin/bash
```

```
fruit="Kiwi"
```

```
case $fruit in
```

```
"apple") echo "Apple pie is quite tasty."
```

```
;;
```

```
"banana") echo "I like banana nut bread."
```

```
;;
```

```
"Kiwi") echo "New Zealand is famous for kiwi."
```

```
;;
```

```
esac
```

Example 2

```
#!/bin/bash

clear
echo "This script tells the use of case command"

echo "1 : Print the Date"
echo "2 : Print the current working directory"
echo "3 : Exit"

echo -n "Enter your choice : "
read reply

echo

case $reply in

"1") date
;;
"2") pwd
;;
"3") exit
;;
*)
echo "Illegal Choice"
;;
esac
```

II. To develop a Basic Math calculator Using case Statement.

Implementation of Calculator application

```
#!/bin/bash
```

```
j=1
```

```
while [ $j -eq 1 ]
```

```
do
```

```
echo "Enter the First Operand;"
```

```
read f1
```

```
echo "Enter the second operand:"
```

```
read f2
```

```
echo "1-> Addition"
```

```
echo "2-> Subtraction"
echo "3-> Multiplication"
echo "4-> Division"

echo "Enter your choice"
read n
case "$n" in

1) echo "Addition"
f3=$((f1+f2))
echo "The result is:$f3"
;;

2) echo "Subtraction"
let "f4=$f1 - $f2"
echo "The result is:$f4"
;;

3) echo "Multiplication"
let "f5=$f1 * $f2"
echo "The result is:$f5"
;;

4) echo "Division"
let "f6=$f1 / $f2"
echo "The result is:$f6"
;;

esac

echo " Do you want to continue(press:1 otherwise press any key to quit)"
read j

done
```

TASK

1. Write a Bash script that takes a username as an argument and prints the user's ID, home directory, and login shell using switch case.
2. Write a Bash script that takes three arguments: two numbers and an operator (+, -, *, /). The script should perform the calculation and print the result handle by using switch case .
3. Write a Bash script that counts and prints the number of files in a given directory and its subdirectories handle by using switch case.